The Belle LLC

Higgins Waterfront Mixed Use

IMEG #26000379.00

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Basis of Design Narrative
for
Higgins Waterfront Mixed Use
Missoula, MT

IMEG #26000379.00
March 13, 2026

A. Introduction

1. Purpose

a. The purpose of this report is to describe the proposed building systems for design coordination, preliminary pricing, and Owner review.

b. The proposed design assumptions are outlined for Owner review and approval. These assumptions include, but are not limited to, external and internal temperature and humidity criteria, noise limits, occupancy, vibration criteria, future expansion needs, fire resistance, and lighting levels.

c. The proposed building systems may evolve in future design phases. Any preliminary pricing must include cost contingencies appropriate to the current phase.

d. Refer to Schematic Design conceptual drawings, project program, and Owner's design and construction standards for additional information.

2. Project Description

a. Higgins Waterfront consists of two 4-story building structures over a common podium garage. The total square footage is 356890 SF. The east building will be a condo building with amenities, first-level townhomes, and condo-style apartments above, totaling 64 units. The west building will consist of two restaurants and a 4-story hotel with guestrooms and amenities, and the hotel brand will be Graduate Hotel by Hilton. The parking garage will be 115,000 sf, with over 290 parking spaces, and will be mechanically ventilated. The two buildings will be provided with separate utilities and MEP systems.

3. Sustainability and Energy Goals

a. This project is not pursuing any sustainability certification programs.

B. Adopted Codes and Standards

1. Codes

    a. Building Code - 2021 International Building Code
    b. Existing Building Code - 2021 International Existing Building Code
    c. Life Safety Code NFPA 101 - 2021
    d. Electrical Code - National Electrical Code NFPA 70 2020
    e. Mechanical Code - 2021 International Mechanical Code
    f. Plumbing Code - 2021 Uniform Plumbing Code
    g. Energy Conservation Code - ASHRAE 90.1 2019 via International Energy Conservation Code - IECC 2021
    h. Fire Protection Code - 2021 International Fire Code
    i. Fire Alarm Code - National Fire Alarm and Signaling Code NFPA 72 2019
    j. Fuel Gas Code - 2021 International Fuel Gas Code
    k. State Department of Labor Requirements
    l. State Department of Health Requirements
    m. State and Federal Safety and Health Laws

C. Owner Standards

1. Published Standards of future Hospitality basis of design criteria will be incorporated and will be based on the Graduate Hotel by Hilton.

D. General Building Design Criteria and Assumptions

1. Summer Space Environmental Requirements: 75°F ± 2°F, 60%RH high limit.
2. Cooling Design Outdoor Air Conditions based on ASHRAE 0.4%
3. Winter Space Environmental Requirements: 70°F ± 2°F 30%RH low limit.
4. Ventilation (Heating Coil) Design Outdoor Air Conditions: based on ASHRAE 99.6% Heating DB
5. Occupant Diversity Factor: no occupancy changes from existing
6. Heating Safety Factor: 15%
7. Cooling Safety Factor: 10%
8. Typical Hours of Operation: 24/7
9. Envelope Assumptions: MT state minimum.
10. Envelope Assumptions:

	U-value (Btu/h*sf*F)	SHGF
Wall		NA
Roof		NA
Glazing (35% wall area)		
Skylight		

Note 1: Center of glazing U-value

E. Mechanical Design Requirements

1. General HVAC System Description

a. The condominium building will consist of a split system for each condo and multiple split systems for corner condos with two thermally preforming zones and facades facing two different cardinal directions. The condo units will have unitized ERVs for ventilation and horizontally ducted fan coils for heating and cooling of spaces.
b. The hotel building will consist of a VRF system throughout the building for heating and cooling of guest rooms, the restaurants, and all amenities. Localized ERVs will be used for amenity ventilation with Demand Control Ventilation (DCV) as required by code while guestrooms will be ventilated with local exhaust fans utilizing common shafts between adjacent guestrooms up to the roof, and packaged rooftop terminal units (RTUs) for providing fresh air to the guestrooms. Guestroom ventilation will be controlled with motorized dampers to close off ventilation to unoccupied guest rooms.

2. Air Handling System

a. Air Handling System Type
    1) Constant volume
b. Air Handling Equipment
    1) Air Handling Unit
        a) Pre-manufactured air handling rooftop terminal units will be used to supply conditioning and ventilation air to the corridor spaces for both buildings. The units will be installed on the roof in the mechanical screened area. These units will be gas-fired with DX cooling coils.
        b) Pre-filters will be provided upstream of all coils. Final filters will be provided downstream of supply fans.
        c) Outside air capability will include a minimum-position damper with a separate economizer damper to provide ventilation and free cooling.

d) The outside air section will contain a flow station to monitor the quantity of outside air introduced.
e) Return fan(s) will be double width double inlet (DWDI) vane-axial plenum type. Blades will be airfoil backward-inclined forward-curved type.
f) Supply fan(s) will be DWDI vane-axial plenum type. Blades will be airfoil backward-inclined forward-curved type.
g) Exhaust fan(s) will be DWDI plenum type. Blades will be airfoil backward-inclined forward-curved type.
h) Motors will be electronically commutated (ECM) AC induction with VFDs.
i) The unit will be provided with preheat electric coils.

2) Packaged Rooftop Air Handling Unit
   a) A packaged rooftop air handling unit will be used to supply conditioning and ventilation air to top floor amenity and restaurant spaces.
   b) The unit will contain a maintenance/service vestibule to house all piping and control valves.
   c) Motors will be electronically commutated (ECM) AC induction with VFDs.
   d) The unit will be provided with preheat electric coils.
   e) The unit will be provided with a sensible-only sensible plus latent energy recovery wheel between the outside and relief exhaust airstreams.
   f) Pre-filters will be provided upstream of all coils. Final filters will be provided downstream of supply fans.
   g) Outside air capability will include a minimum-position damper with a separate economizer damper to provide ventilation and free cooling. The outside air section will contain a flow station to monitor the quantity of outside air introduced
3) Ventilation System
   a) Exhaust System
   b) General exhaust will be provided for all restrooms, showers, and janitor’s closets. Separate exhaust fans will be provided for the grease-laden kitchen hood and dishwasher exhaust.
   c) General exhaust ductwork will be galvanized steel and will be insulated within 10’ of roof penetrations with fiberglass insulation wrap. Moisture-laden exhaust ductwork from showers will be aluminum.

d) Garage exhaust will be dual exhaust fans from the lowest level of the garage, pulling in make-up air from the garage entry door. Air in the garage will be monitored by NO2/CO sensors to ramp up mixing fans and exhaust fans as needed.

F. General Plumbing Description

1. Plumbing systems shall consist of all fixtures, potable cold and hot water piping and equipment, piping insulation and all sanitary waste and vent piping systems, grease waste systems, roof and storm drainage systems, gas piping systems. All plumbing sys¬tems shall be designed and constructed in accordance with the 2021 Uniform Plumbing Code with Montana amendments.

1. Plumbing Fixtures

1) Condos & Town Homes

a. Plumbing fixtures in the building shall be residential-grade equal to Kohler, based on the owners' requested fixtures. All fixtures to be non-hands free type.
b. Water closets shall be floor-mounted vitreous china flush tank toilets and ADA-compliant where required.
c. Lavatories shall be vitreous china, countertop type, or under-mount with a single-lever faucet.
d. Showers and shower/tub combinations shall be fiberglass with single-lever faucets and shower heads. ADA-compliant showers will be provided in appropriate apartments and shall have a handheld shower head.
e. Sinks will be stainless steel by Just, Elkay or other approved manufacturer. Fixture trim will be a single-handle faucet with a pull-down spray. Dishwashers and disposals will be provided by the architect and installed by the contractor. Ice maker connection for refrigerators will be provided.
f. Washing machine supply and drain connections will be provided.
g. Hub drain and or riser will be provided for HVAC condensate and water heater drain. Condensate routed to sanitary.

2) Hotel Guest Rooms

a. Plumbing fixtures shall be residential-grade equal to Kohler, based on the owners' requested fixtures and hotel brand standards. All fixtures to be non-hand-free type and sensor-operated.
b. Water closets shall be floor-mounted vitreous China flush tank toilets and ADA-compliant where required.
c. Lavatories shall be vitreous China, countertop, or under-mount type with a single-lever faucet.

d. Showers and shower/tub combinations shall be fiberglass with single-lever faucets and shower heads. ADA-compliant showers will be provided in appropriate apartments and shall have a handheld shower head.

3) Hotel Retail, Food Service & Laundry
   a. Plumbing fixtures, lavatories, wall-mounted water closets, and urinals will be porcelain type. Fixture colors will be selected by the Owner Architect. Sensor-operated manual faucets, sensor-operated water closets, and urinals will be provided for public restrooms.
   b. Floor drains are required for toilet rooms with urinals, toilet rooms with more than one water closet, toilet rooms serving food service areas, including kitchens, mechanical/plumbing rooms, and fire protection riser rooms.
   c. The laundry will be provided with a lint trap serving the washing machines.
   d. Kitchens, fixtures and faucets will be provided by the food service consultant. Plumbing contractor will provide piping & piping accessories, floor sinks, floor drains and gas piping.
   e. Grease interceptors will be provided for grease laden fixtures and equipment. Grease interceptors will be installed in the ground outside of the building footprint.

4) Domestic Water Systems
   a. Water Service Connections will be provided as follows:
      1) Hotel: approximate size 4”. A triplex booster pump is required to serve the guest rooms and all other hotel spaces. Two pressure zones will be required, street pressure serving levels P- 2 and below and boosted pressure serving levels 1 and above.
      2) Town Homes: approximate size 3”. Street pressure is adequate to serve the townhomes.
      3) Condos: approximate size 4”. A triplex booster pump is required to serve the condominiums.
   b. Domestic water service entrances will be provided from the City water main. Double-check that reduced-pressure principle backflow preventers will be located in meter pits outside of the building(s) by civil.

5) Water Main Pressures:
   a. Recent water flow test indicates approximately 40 psi for residual flowing 2,000 gpm.

6) Domestic Water Heating Systems:

a. Condos and Town Homes:

7) Water Heater:
   a. Water heating shall be a low-boy 40 or 50-gallon electric water heater and will be located in closets in each unit. One bedroom units to include 40 gallon heaters and two bedroom units to include 50 gallon units.

8) Hotel, Guestrooms, Retail, Food Service & Laundry:
   a. Instantaneous natural gas-fired water heaters will serve the hotel, food service, laundry, and retail spaces. Installed in a plumbing and mechanical room.
   b. A dedicated hot water supply and circulating system at 120 degrees to serve the guest rooms. A master digital electronic thermostatic mixing valve will control the temperature.
   c. A dedicated hot water supply and circulating system at 140 degrees will serve the food service and laundry areas. The water heater thermostats will control the 140-degree temperature.
   d. Local at the fixture thermostatic mixing valves will be installed on all public hand washing lavatories set to 105 degrees.

9) Recirculation Systems:
   a. Fractional-horsepower circulating pumps will be provided to maintain hot water temperatures for all systems.
   b. Balancing valves and check valves will be provided to control water flow throughout all systems. Pump controls shall meet IECC requirements.

10) Sanitary Drain, Waste and Vent Systems:
   a. Water Service Connections will be provided as follows:
      1. Hotel: approximate size 8", may be oversized to afford less pipe slope of 1/16th inch per foot.
      2. Town Homes: approximate size 8", may be oversized to afford less pipe slope of 1/16th inch per foot.
      3. Condos: approximate size 8", may be oversized to afford less pipe slope of 1/16th inch per foot.
   b. Sanitary service entrances will be provided by the City.
   c. Lift stations will be provided in the bottom level of the parking garage. One unit serves the hotel, one unit serves the townhomes. The lift stations shall be duplex pumps with controls and control panels. Traffic rated access covers airtight.

d. Grease interceptors shall be provided, two are required serving the hotel food service areas. Grease interceptors shall be underground with traffic rated access covers.

11) Storm and Roof Drainage Systems:
   a. Numerous connections will be required around the facility. Civil will collect all and route to the point of disposal.
   b. Roof drainage systems shall include a secondary drainage system.
   c. Courtyard is not covered and is located entirely over a structural slab over the underground parking garage. Plumbing will provide the drainage of the courtyard.
   d. The uncovered courtyard areas shall be provided with drains routed to civil connections. A trench drain shall be provided at the entrance to the parking garage.
   e. Lift station will be provided for the covered garage areas. The lift station shall be a duplex pump with controls and control panels. Traffic-rated access covers airtight. Covered garage discharge from the lift station shall connect to the sanitary drainage system.
   f. Floor drains are required throughout the parking garage.
   g. An oil water separator and sand trap are required for the treatment of the flow prior to entering the lift station.

12) Natural Gas Systems:
   a. Three separate systems will be required for the facility. One system to serve the Hotel, one system to serve the Town Homes, and one system to serve the Condos. Civil will show the connections to the utility gas system.
   b. Meters and regulators will be provided and installed by the gas utility provider.

13) Piping Systems:
   a. Domestic Water:
      1. Domestic water supply piping systems shall be constructed of Type L Copper, CPVC piping up to 2, CPVC piping 2.5” and larger shall be Sched-ule 80,PEX is approved for the apartment units and guestrooms, manifolds are acceptable, or stainless steel systems.
      2. Shutoff valves will be located at all major branch and all final use locations.

b. Sanitary Waste and Vent System Description:
    1. Sanitary waste and vent systems shall be constructed of Cast Iron No Hub, PVC DWV sch 40 solid core or DWV Copper.
    2. Cleanouts shall be provided in accordance with local code.
    3. Cast Iron piping is required for roof drainage piping located in areas where sound is sensitive.

c. Storm and Roof Drainage Systems:
    1. Storm and roof drainage systems shall be constructed of Cast Iron No Hub or PVC DWV Sch 40 solid core.
    2. Cleanouts shall be provided in accordance with local code.
    3. Cast Iron piping is required for roof drainage piping located in areas where sound is sensitive.
    4. Dual, primary and secondary, roof drains will be required. Downspout nozzles will be provided for secondary drainage terminals.

d. Natural Gas Systems:
    1. Natural gas systems shall be constructed of Sch 40 black iron pipe. Connections to equipment shall be stainless steel corrugated.
    2. Shut off valves, unions, secondary regulators and drip legs shall be provided at all connections to equipment and appliances.

14) Pipe Insulation:

a. Provide fiberglass or elastomeric flexible rubber pipe insulation, provide PVC jacket where exposed, as follows:
    1. Hotel:
        a) All domestic cold, hot and hot water return piping.
        b) All horizontal roof/storm piping.
        c) Exposed pipes located in knee spaces of A.D.A. fixtures.
        d) Provide electric heat trace with insulation for piping exposed to freezing temperatures.
    2. Condos and Town Homes:
        a) All domestic hot water supply and return piping.
        b) Exposed pipes located in knee spaces of A.D.A. fixtures.
        c) Provide electric heat trace with insulation for piping exposed to freezing temperatures.
    3. Parking Garage:

a) Provide electric heat trace with insulation for piping exposed to freezing temperatures.

4. Elevator Sump Pumps:
   a) Provide sump pumps in all elevators, pumps to include oil waste controls to prevent oil from entering the discharge.
   b) Pumps are required to provide 50 gpm per elevator car served.

G. Electrical Design Requirements

1. Overview:
   a. This section describes the proposed electrical infrastructure serving the development. Electrical systems will be designed to support the operational needs of the hotel, residential, retail, and shared building areas while maintaining separation between hotel and residential electrical infrastructure where practical.
   b. Two primary electrical services are anticipated for the project: one serving the residential portion of the development and a second serving the hotel and associated commercial spaces. This dual-service approach allows the differing voltage and operational requirements of the two building uses to be accommodated efficiently.
   c. All electrical service sizes and distribution concepts described in this narrative are preliminary and are based on information available during the Schematic Design phase. Final service sizing, equipment selection, and distribution strategies will be refined as mechanical loads, elevator equipment, fire protection systems, and tenant requirements are further defined.

2. Utility Service and Electrical Infrastructure
   a. Electrical service for the development will be provided by Northwestern Energy. Current planning assumes two separate utility transformers serving the project.
   b. The residential portion of the development is anticipated to be served by an approximately 3000 ampere, 208Y/120V, three-phase electrical service. A dedicated residential electrical service entrance room will house the service equipment and distribution gear serving the residential systems.
   c. The hotel portion of the development is anticipated to be served by an approximately 2000 ampere, 480Y/277V, three-phase electrical service. A dedicated hotel electrical room will contain the primary service switchboard and associated distribution equipment.
   d. Separate pad-mounted utility transformers are anticipated for the hotel and residential services. Final transformer locations will be coordinated with the utility provider, the design team, and site planning requirements.

e. Final electrical service sizes will be verified as part of the Design Development phase once detailed mechanical equipment schedules, elevator loads, EV charging requirements, and other major electrical loads are finalized.

3. Residential Electrical Distribution

   a. The residential portion of the development currently includes approximately 63 residential units consisting of a mixture of condominium and townhome-style units.
   b. Each residential unit will be individually metered through utility-owned meters located within a dedicated residential meter room. Electrical feeders will extend from the meter room to the respective residential units via a coordinated vertical electrical riser system.
   c. For preliminary planning purposes, residential unit panelboard sizes are assumed as follows:
      1) One-bedroom residential units: 100A, 120/208V single-phase panels
      2) Two-bedroom residential units: 125A, 120/208V single-phase panels
      3) Three-bedroom and larger residential units: 150A, 120/208V single-phase panels
   a. A vertical electrical chase will be provided to accommodate the residential feeder riser system serving the upper residential floors. Final sizing and routing of the electrical riser will be coordinated with architectural and structural systems as the building design progresses.
   b. Common residential area loads, including corridors, lobbies, and residential support spaces, will be served from residential house panels located within electrical rooms serving the residential portion of the building.

4. Hotel Electrical Distribution

   a. The hotel electrical system will be based on a 480Y/277V, three-phase service entering a main hotel electrical room containing the primary service switchboard.
   b. From the main switchboard, power will be distributed to mechanical equipment, elevator systems, and other large building loads operating at 480 volts. Step-down transformers will be provided throughout the building to generate 208Y/120V distribution systems serving receptacles, small equipment loads, and guestroom power requirements.
   c. Electrical distribution panels and transformers will be located in dedicated electrical rooms on each floor as required to support efficient distribution and maintain

manageable feeder lengths. The final electrical room locations and sizes will be coordinated with the architectural design as the project progresses.

d. Retail tenant spaces located within the project will be served from the hotel electrical infrastructure. Retail tenants are currently anticipated to be provided with individual utility meters derived from the hotel electrical transformer. Final tenant electrical requirements will be coordinated during tenant fit-out.

5. Shared Podium and Garage Systems

   a. The development includes a shared parking garage and podium structure serving both the hotel and residential portions of the project.
   b. Electrical ownership and service allocation for shared systems within the garage and podium have not yet been finalized and will require direction from the Owner. These shared loads may include garage lighting, garage ventilation systems, EV charging equipment, and site lighting.
   c. For design efficiency, it is recommended that shared lighting systems within the garage and exterior site lighting be served from the hotel electrical service where feasible. The 480Y/277V electrical system associated with the hotel allows efficient distribution of 277-volt lighting circuits, reducing the number of transformers and improving overall system efficiency.
   d. Final assignment of shared electrical loads between the hotel and residential services will be confirmed during later phases of design in coordination with the Owner.

6. Fire Protection Electrical Systems

   a. Current planning assumptions include separate fire protection systems serving the hotel and residential portions of the project. Each building is anticipated to include a dedicated fire pump.
   b. Electrical power for each fire pump will be provided from the respective building service equipment through a dedicated feeder connection ahead of the service disconnecting means, in accordance with applicable electrical codes.
   c. Final fire pump sizes, electrical characteristics, and any standby power requirements will be verified in coordination with the fire protection engineer, mechanical engineer, and the Authority Having Jurisdiction as the project progresses.

7. EV Charging Infrastructure

    a. Electric vehicle charging stations are anticipated within the parking garage as part of the project program. The quantity of EV charging stations and their final electrical service allocation are currently under development.
    b. The local Fire Marshal has requested that EV charging stations within the garage be located near building exits. This requirement will influence the routing of electrical distribution and conduit infrastructure serving the EV charging equipment.
    c. Final EV charger quantities, electrical load requirements, and service assignments will be confirmed as the project develops.

8. Telecommunications and Low Voltage Infrastructure

    a. Dedicated telecommunications entrance facilities will be provided for both the residential and hotel portions of the development. These spaces will accommodate telecommunications service provider equipment and serve as the demarcation point between external telecommunications infrastructure and the building distribution systems.
    b. Electrical design will provide power connections and conduit infrastructure supporting telecommunications and low-voltage systems. Detailed design of telecommunications, data, security, and other low-voltage systems will be developed by the project technology design team.

9. Electrical Space Planning

    a. The project will include several dedicated electrical spaces to support the electrical infrastructure of the development. These spaces include the residential electrical service entrance room, residential meter room, residential telecommunications demarcation room, and hotel main electrical room.
    b. Additional secondary electrical rooms will be distributed throughout the building to support local power distribution to residential floors, hotel guestroom floors, and other building areas. Preliminary planning assumes secondary electrical rooms on upper floors in the range of approximately 100 to 150 square feet, though final sizing will depend on final equipment layouts.
    c. All electrical room sizes and locations remain preliminary and will be refined as electrical equipment layouts are developed during later design phases.

10. Design Assumptions and Items Requiring Verification

    a. Several project elements remain under development and will require further coordination during the Design Development phase. These items include confirmation of final electrical service sizes, allocation of shared garage and site electrical loads between the hotel and residential services, final fire pump requirements and associated electrical loads, final elevator equipment loads, and the quantity and electrical requirements of EV charging infrastructure.
    b. Additional coordination will also be required to confirm mechanical equipment electrical loads, final lighting distribution strategies, and detailed telecommunications infrastructure requirements.
    c. As the project design progresses and additional information becomes available, the electrical design will be refined to accommodate final equipment selections, code requirements, and Owner preferences.

11. Lighting Systems and Design Assumptions

    a. Overview:

        1) The electrical design for this project will include the components of lighting systems and controls, power distribution, fire alarm and automatic detection.

        2) Refer to the Technology Design Requirements for a description of the remaining electrical-based low-voltage system requirements.

    b. Lighting Systems and Controls

    c. Lighting System Design Criteria

        1. Lighting basis of design per the maintained illumination targets published by the Illumination Engineering Society (IES), applicable to local municipal lighting ordinance, and Owner standards. Refer to the average maintained illuminance level criteria in the table below.
    d. Correlated Color Temperature (CCT)
        1. Interior Luminaires: 3000 °K
        2. Exterior Luminaires: 3000 °K
    e. Color Rendering Index (CRI)
        1. Interior Luminaires: 90
        2. Exterior Luminaires: 70 or 80

f. Interior Luminaires
   1. Specification-grade LED source luminaires with switched or dimmable drivers, minimum 50,000 hour rated life.
   2. Luminaires recessed in the thermal envelope will be IC rated and airtight.
   3. Special Application Listings: Listed wet, damp, or ingress protection IP## rated, Hazardous Class # - Division # listed by application.

g. Exterior Luminaires
   1. Specification-grade LED-based light source luminaires with dimmed drivers, minimum 50,000-hour rated life.
   2. LED drivers, low temperature, reliable start at -20°F (-29°C) ambient, reliable operation without derating at 140°F (60°C).
   3. Luminaire must POWER DOWN during daytime hours in addition to dimming to zero output. All additional heat gains shall be avoided.
   4. Dark Sky Lighting Zone: LZ1Low ambient lighting to LZ2 Moderate ambient lighting.
   5. Dark Sky Compliance: Full cutoff.
   6. Listed wet, damp, or ingress protection IP## rated.

h. Egress and Emergency Egress Lighting Design Basis
   1. Egress (non-emergency) lighting will be provided for all corridors, stairways, and egress paths defined by the Architect.
   2. Emergency egress lighting will be served from central lighting inverters for most publicly accessible spaces and dedicated battery-powered wall/ceiling-mounted luminaires with self-diagnostics and testing where connection to a central lighting inverter is not feasible.
   3. Exterior Entrance / Discharge: Provide a minimum of two LED sources and drivers for normal and emergency egress code compliance.
   4. Exit Signage
      a) Specification grade LED type, diecast edge lit served from central lighting inverter in publicly accessible areas. Exit signs will include integral emergency battery with self-diagnostics and testing in areas where connection to a central lighting inverter is not feasible.
      b) Ceiling / wall-mounted for all applicable egress paths, exterior exits, assembly spaces, and code-required spaces.
      c) Wall mounted (at floor) for egress marking applications required by adopted code.
   5. Automatic load control relays (ALCR) will be provided for each lighting control zone when served by an emergency power source remote from the luminaire.

12. Lighting Control System Criteria
    1) Additional mandatory controls for lighting that include manual switching/dimming, automatic controls to reduce lighting levels, and daylight responsive controls will be installed.
    2) Lighting Control System Basis of Design
        1. Stand Alone and Room Controllers: Acuity Controls Legrand Watt Stopper Hubbell Automation Eaton Greengate Lutron
        2. Network Controllers: Electronic Theater Controls ETC Lehigh Lutron
    3) The energy code compliant lighting control system will include a combination of standalone controls, network-based controls. Refer to the Lighting and Lighting Control System - Space Description Table. The following system components will be included:
        1. Wall Mounted: Switches, dimmers, scene selection control stations, manual override stations.
        2. PIR, Ultrasonic, and Combination Sensors: Occupancy and vacancy based.
            a) Where allowed, a vacancy sensor that controls lighting with a dimmable driver will come on automatically to 50% and manually to 100%.
        3. Time Clock: Astronomical
        4. Photocell
        5. Central Programming and Scheduling: 5/7 day, holiday, special event. Remote online monitoring, programming, and override secure access.
        6. Communication / Protocol: Combination of line voltage, and low-voltage wireless.
    4) Daylight Responsive Control

1. Daylight responsive controls will be used in areas required by the energy code. In addition, the following spaces provide a value-added energy savings opportunity for daylight responsive control.
    a) Additional Spaces: Public areas
2. Dimming control will be used to modulate artificial lighting sources up/down based on available daylighting contributions.
    5) Receptacle Control
        1. Receptacle control will be used in areas required per the applicable energy code. Applicable energy codes require fifty percent of the wiring device receptacles to be controlled with the luminaires when the space has been unoccupied for predetermined quantity of time. Refer to Wiring Devices for specification of controlled receptacles.
        2. Receptacles will be automatically controlled with occupancy sensor or timeclock control with override switch.

3. Communication will be hardwired or wireless with compatible control devices.

Lighting and Lighting Control System - Space Description Table			
Space Description	Luminaires	Lighting Controls	Illuminance Levels
Amenity Spaces			
Lobby, Reception	Recessed downlighting and architectural accent luminaires, attention to accent and vertical illumination	Networked lighting control dimming system with occupancy sensor and control Dimming daylighting control	10 to 20 foot-candles
Vestibule, Main Entrance	Recessed downlights Luminaires, accent per Interior Design team Indirect lighting at underside of Porte Cochere	Networked lighting control dimming system with occupancy sensor and control Dimming daylighting control Fire alarm override	10 to 15 foot-candles
Public Restrooms and Locker Rooms	Recessed downlight, at vanity and in in shower Architectural accent luminaires per Interior Design team Emergency lighting per code	Networked lighting control dimming system with Occupancy sensor and control Dimming daylighting control	General: 5 foot-candles Vanity: 20 foot-candles
Single Use Restrooms	Recessed downlight, at vanity and in in shower Architectural accent luminaires per Interior Design team	Occupancy sensors	General: 5 foot-candles Vanity: 20 foot-candles
Public Corridors, Circulation, Elevator Lobby	Recessed downlighting and architectural accent luminaires per Interior Desing team Emergency lighting per code	Networked lighting control dimming system with Occupancy sensor and control Dimming daylighting control	7 to 10 foot-candles

Stairs	Wall-mounted acrylic stair luminaires mounted on each stair landing	Integral occupancy sensor with minimum quantity of night lighting for entrance, security Fire alarm override	10 foot-candles, minimum on stair treads 15 to 20 foot-candle average during use
Common Workspaces, Residence Support	Linear LED dimmable volumetric type luminaires	Vacancy sensors with wall mounted controller Receptacle control	30 to 40 foot-candles
Dwelling Units			
Dwelling Corridors	Recessed downlighting, linear accent tape light and architectural accent luminaires per Interior Desing team Emergency lighting per code	Networked lighting control dimming system with Occupancy sensor and control Dimming daylighting control	7 to 10 foot-candles
Dwelling Units	Recessed downlighting and architectural accent luminaires per Interior Desing team	Programable dimming system with multi scene presets	General; 7 foot-candle Bed for Reading: 20 foot-candles Task light at desk
Dwelling Toilet	Recessed downlight, at vanity and in in shower Architectural accent luminaires per Interior Design team	Programable dimming system with multi scene presets	General: 5 foot-candles Vanity: 20 foot-candles
Dwelling Kitchen	Recessed downlighting and architectural accent luminaires per Interior Desing team	Programable dimming system with multi scene presets	General: 10 foot-candles Counter Tops: 30 foot-candles
Exterior Lighting			
Building Facade	Linear LED wash and grazing, recessed accent, architectural accent luminaires per Interior Desing team Emergency lighting per code	Networked lighting control dimming system with Vacancy sensor and control Dimming daylighting control Networked lighting control dimming system with Vacancy sensor and control Dimming daylighting control	Per design, Dark Sky compliant

Pedestrian Walkways, Bike Storage, and Sidewalks	Exterior building mounted luminaires for building adjacent spaces Site pole mounted and architectural bollards for spaces remote from the building	Networked lighting control dimming system with Vacancy sensor and control Dimming daylighting control	0.5 to 1 foot-candle at walkways 3 to 5 foot-candle at main intersections
Entry / Exit, Secondary	Building-mounted luminaires Architectural accent luminaires per Interior Desing team	Networked lighting control dimming system Dimming daylighting control	1 to 3 foot-candle
Exterior Amenity spaces	Architectural accent luminaires per Interior Desing team, Emergency lighting per code	Networked lighting control dimming system with Vacancy sensor and control Dimming daylighting control	7 to 10 foot-candles
Monument Signage	Self-illumination provided by signage vendor or Architectural landscape lighting	Astronomical time clock and photocell control	Varies, aesthetic
Flagpole Monument	In-grade flagpole luminaire	Astronomical time clock and photocell control	Varies, aesthetic
Architectural Landscape	Ground mounted landscape lighting and in-grade specification grade architectural landscape lighting	Networked lighting control dimming system with Vacancy sensor and control	Varies, aesthetic

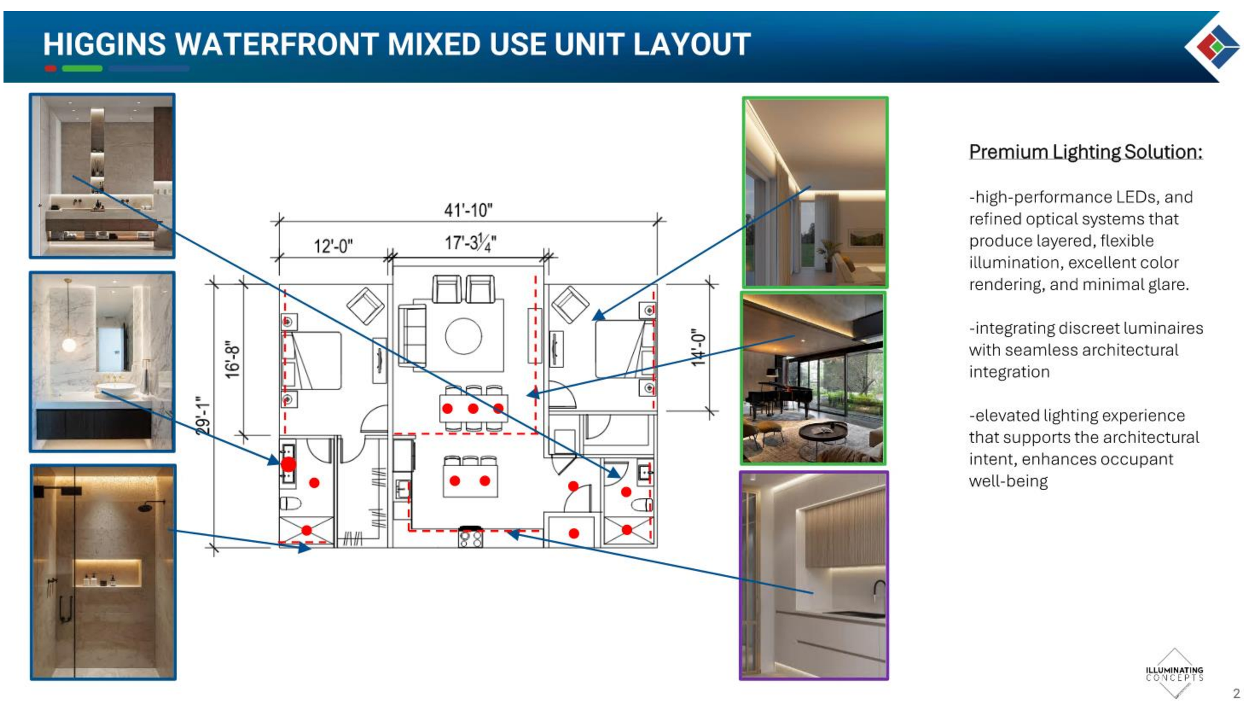
Basis of Design Luminaire Schedule		
Luminaire Type	**Description**	**Basis of Design Manufacturer / Series Manufacturer**
Recessed Down Light	Lensed downlights with integral LED light modules, flangless trim, trim to match surrounding ceilings	Tech Lighting Entra DMF Lighting spec grade products Alphabet Lighting
Recessed Perimeter Cove Large	Recessed engineered perimeter light wall wash or graze	Prudential Lighting Ledalite
Recessed Direct/Indirect	Indirect LED, steel housing with two curved reflectors and side-mounted LEDs	Prudential Lighting Ledalite
Suspended Direct/Indirect	Rectangular Round linear suspended direct/indirect, steel aluminum housing	Prudential Lighting Ledalite
Architectural Decorative	As per Interior Design Team	Similar to Visual Comfort
Task Light, Under Cabinet	Under cabinet luminaire, diffuse matte	QLT Lighting Precise Lighting Kelvix Lighting
Recessed Acrylic Luminaire	LED static grid lensed troffer,	Acuity Lithonia GTL Series Cooper Metalux GR Series Signify Daybrite TG Series
Recessed volumetric Luminaire	Direct with high angle illuminance	Acuity Lithonia RT Series Cooper Metalux Accord AC Series H.E. Williams HET Series Current Columbia LEPC Series Leviton Viscor LRTJ Series
Exit Sign, Die-Cast	Emergency exit sign, white black die-cast aluminum body, red green letters, universal arrows/mounting.	Hubbell Dual-Lite EVE Series Acuity Lithonia LQMS Series

	emergency ni-cad battery self-test and diagnostics of inverter and lamps	Signify Chloride CLX Series Cooper Surelites APX Series
Exit Sign, Edge-Lit	Exit sign, injection molded acrylic mirror lens and extruded aluminum housing, edge lit, red letters, clear background, Verify recessed end, back or ceiling mounting and arrows with plans	Acuity Lithonia LRP Series Hubbell Dual-Lite Le Series Signify Chloride 44r Series
Façade Lighting	Linear LED in IP68 extrusions	QLT Lighting KKDC Lighting Kelvix Lighting
Site Pole	Decorative site luminaire, aluminum extruded housing, gasketed, distribution type to be determined, listed wet location Ip68 rated Pole: aluminum pole with internal vibration damper, anchor or embedded base	Landscape Forms
Landscape Lighting	Specification grade optical system, listed wet location Ip69 rated Concrete base	BK Lighting HK Lighting

HIGGINS WATERFRONT MIXED USE

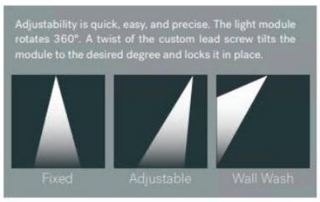
Schematic Design Narrative Dwelling Units

MARCH 2026

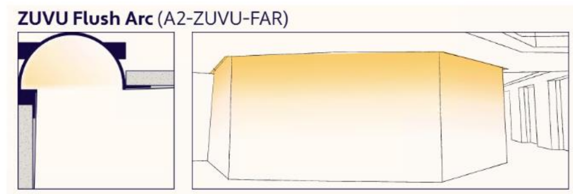


ARCHITECTURAL LIGHTING

Downlight:
-3" used throughout



Integrated Luminaires:
-Reduces the need to build out a formal cove.
-Reduces the installation complexity



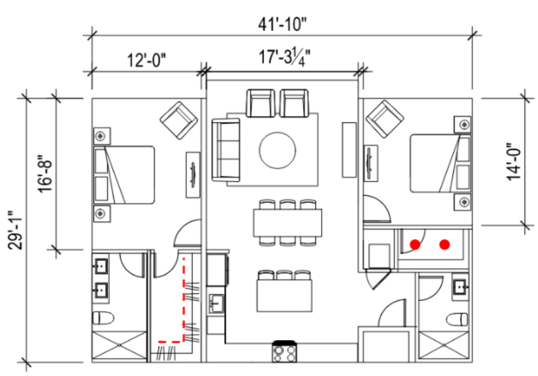
*All decorative fixtures to be spec'd by interior design team

3

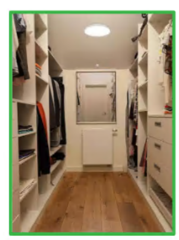
CLOSETS

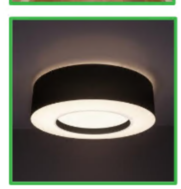
Closet Lighting Essentials:
-Shadow control: high angle illumination is key as it provides an evenly distributed glow
-Task lighting: Under cabinet lighting is essential in these areas

Recessed linear with a drop lens



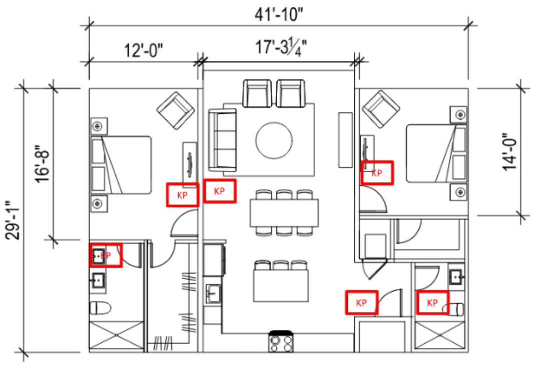
Surface Mount





4

HIGGINS WATERFRONT MIXED USE UNIT LIGHTING CONTROLS



Advanced Lighting Controls:
-Provides user with personalized control of lighting

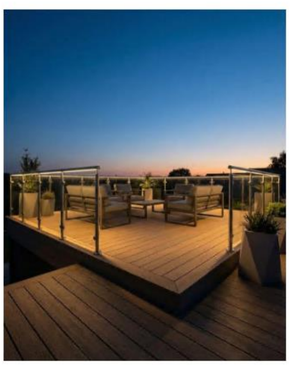
-Scene based: Can be programmed for scenes “Relax”, “Cooking”, etc., which will turn each zone on to a preset dimming setting to create the ideal light levels for any time of day

-Zone based: Can be programmed for specific zones “Downlights”, “Undercabinet”, etc., to remove the confusion of multiple switches

ILLUMINATING CONCEPTS

5

EXTERIORS



Handrail lighting:
-Illuminates the perimeter of the space, a glowing outline exaggerates the footprint
-Extends the usable space at night, people naturally occupy the full footprint

Barbecue Lighting:
-Cylinders to provide task lighting
-Wall grazing to provide an ambient glow
-Toe Kick lighting for ambience and safety

ILLUMINATING CONCEPTS

6

H. Technology Design Requirements

1. Requirements Distributed Antenna System (DAS) For Public Safety Networks
   a. The need for the DAS system will be determined once construction has been completed and the building is tested by a DAS system engineer who will provide a full report for review by the local AHJ. The pathways for the DAS system will be provided in construction.
2. Closed Circuit Television System (CCTV):
   a. A new IP based CCTV camera system will be installed to include camera licenses to operate on a new CCTV systems server. New cabling will be provided for each camera. The camera layout and field of views will be coordinated with the owner and verified prior to installation of the camera.
3. Card Access System:
   a. A new Card Access control system with card readers and associated devices will be installed on all non-unit exterior doors to integrate with wireless locksets provided by the door hardware consultant.
   b. The Tennent vehicle entry points to have a card reader for gate control and Intercom/video entry system for visitors to be connected to the Tenants via smartphone app.
   c. Intercom/video entry system will be provided at all primary entry points.

4. Voice/Data/CATV Structured Cabling:
   a. Backbone cabling to new Technology rooms will consist of a minimum of one 12-strand OS2 single mode fiber optic data cable as required if more than one telecommunication room is provided, and one (1) .500 hardline coaxial copper CATV cable as required to support the new building.
   b. Horizontal station cabling will consist of Category 6 UTP copper voice and data cabling and RG-6 quad-shielded coaxial copper CATV cabling, fed from the nearest Technology room to non-unit locations. Where installed CATV cable is longer than 180 feet, RG-11 quad-shielded coaxial copper CATV cables will be installed in lieu of the aforementioned RG-6.
   c. Structured Cabling systems will be installed to support Phone systems, POS systems, Security systems, Wireless Access Point system, Data systems, Cellular – DAS system and building automation systems.
   d. Tennant unit horizontal cabling will consist of two Category 6 UTP cables to each unit from the closest telecommunication room. Each unit will consist of a wireless access point location, designated phone location, TV locations and any data locations to be cabled to a central media enclosure with Category 6 UTP.

e. Information outlet location configuration(s) and locations will be determined by Owner's standards for information outlet configurations and locations, equipment that requires structured cabling connection(s). Information outlet locations for wireless access points will be provided, spaced to provide WAP coverage with a 30-foot radius to support 802.11ac connectivity throughout the project area to include outdoor locations in designated areas. Layout of WAP locations will be confirmed by Owner prior to issuance of bid documents. Each WAP location cabling will consist of one Category 6 cable.
f. Cable pathways for all structured cables will consist of conduit and J-hooks as determined by space requirements. Typical information outlet location rough-in will consist of a minimum of one (1) 4 squared back box with single-gang plaster ring and a 1" EMT conduit routed.
g. Technology rooms will be sized to provide adequate space to support all current and foreseeable future needs of the building and will be laid out to provide necessary working clearances for all installed rack-mounted and wall-mounted passive and active equipment.
h. Design will include equipment racks, wall termination space, fiber optic termination cabinets, copper termination patch panels, and full cable management for all cabling, which will include ladder rack for overhead distribution of cabling and equipment patch cords, vertical and horizontal cable management on equipment racks, and D-rings at wall-mounted equipment.
i. Fiber Optic data backbone cabling will be terminated in rack-mounted fiber optic termination cabinets.
j. Voice backbone cabling will be terminated on rack-mounted 110 patch panels.
k. CATV backbone cabling will be terminated directly on wall-mounted CATV distribution equipment. Horizontal voice and data cabling will be terminated in Technology rooms as universal horizontal cabling on the same rack-mounted RJ-45 patch panels. Horizontal CATV cabling will be terminated in Technology rooms directly on wall-mounted directional coupler / taps.
l. The Technology grounding and bonding system will include a wall-mount busbar in each Technology room, rack-mount busbars on each equipment rack, bonding backbone conductors, and equipment, hardware, and device bonding conductors.

5. Area of Rescue Communication
   a. Area of Rescue Communications will be provided at each elevator above and below the main entry floor of each building in the elevator lobbies per building code and AHJ

requirements. The master station will be located in the primary entry vestibule or staffed lobby area.

6. Audio Video Systems
   a. Professional Audio Video System
      1) A full audio/video system will be provided in conference rooms and large commons spaces that will include video display devices (LCDs, plasmas, projectors), a touch screen control system, source equipment and amplification.

7. Acoustics
   a. References
      1) Acoustic criteria are based on IMEG's experience with similar facilities and the following reference standards:
         a) 2023 ASHRAE Handbook for HVAC Applications Chapter 49 Noise and Vibration Control will be followed for recommendations of mechanical noise and to meet the minimum standards of care.
         b) International Building Code (IBC) 2024.
      2) We understand that sustainability certifications (e.g., LEED, WELL, etc.) are not currently being pursued.
   b. Criteria
      1) Exterior MEP Equipment Noise Control
         a) Review is pending, specifically on whether there are exterior requirements applicable to stationary MEP equipment. Additional comments addressing MEP noise impacts to the exterior will be provided in the subsequent design stage.
      2) Environmental Noise
         a) Interior Background Noise – Condominiums
            1. The maximum internal ambient noise levels include contributions from MEP systems, exterior road traffic, and internal noise sources. These maximum sound levels shall be:
               a) Limit road traffic noise intrusion to 45 L_{Aeq}, 1hr for day, 40 L_{Aeq}, 1hr at night.
               b) Short term day/night noise limits for sirens and aircraft is 50 L_{Amax}
               c) 45 L_{dnA} for aircraft noise intrusion

      3) Mock-Ups
         a) Much can be gained from evaluation of a condominium mock-up to provide assurance of the acoustic performance of various issues that lead to complaints.
         b) If the Owner has interest in acoustic testing of mock-up spaces, we recommend that they be planned to allow for the following testing:

1. Demising partition performance, including doors and mullion detail. This will confirm that standard workmanship and all detailing addressed in the construction documents will be sufficient to provide repeatable acoustic performance.
2. Guestroom HVAC options to confirm sound level and quality of various units. If FCUs operating over NC 38 are intended, we recommend conducting a listening mock-up including additional units. Such a test may allow for less-expensive units to be evaluated and accepted by Ownership.
3. Flooring underlayment options. There are many options, and some of the least expensive options can greatly vary in acoustic performance. Mock-up(s) only need to install a section of 4’x4’ finished floor.

c) Additional mock-up(s) for testing may include:
1. Operable partition to assess detailing and acoustic seal performance
2. Slab-edge at façade detail to assess the horizontal and vertical flanking paths
3. Guestroom and condominium connecting doors (if applicable) to assess detailing and acoustic seal performance

d) Mock-ups shall be constructed on-site whenever possible

4) Sound Isolation – Interior

a) Airborne noise is commonly quantified by the Sound Transmission Class (STC) rating, a controlled laboratory test of an assembly. The on-site airborne sound isolation of an assembly is typically reduced due to flanking paths and is commonly quantified by the Apparent Sound Transmission Class (ASTC) rating.

b) Limits for impact noise transmission between vertically adjacent spaces are defined by Impact Insulation Class (IIC). Similar to airborne sound isolation, the on-site impact noise transmission of an assembly is typically reduced due to flanking paths and is commonly quantified by the Apparent Impact Insulation Class (AIIC) rating.

c) The following table provides criteria for sound isolation of demising constructions by room adjacency.

Sound Isolation Recommendations			
Adjacency		Acoustic Criteria (Minimum)	
		STC	IIC
Condominiums	Demising walls	55	-
	Shaft and chase walls	55	-

Sound Isolation Recommendations			
	Elevator shaft walls (double stud)	60	-
	Corridor walls (with door)	50	-
	Vending / Ice machine alcoves	55	-
	Bathroom perimeter walls	54	
	Exterior glazing	TBD*	-
	Floor / ceiling construction	50	55
Retail	Floor / Ceiling construction to condominiums	60	≥65
Private Offices	Demising walls	45	-
	Corridor walls (with doors)	42	-
	Elevator shafts, pantry cabinet walls (double stud)	55	-

*NOTE: Acoustical performance of the exterior façade glazing system should be determined by the condo developer standards whether they prefer to meet minimum code requirements for exterior noise intrusion or if there is a desire to provide enhanced glazing for sound isolation from other exterior activities. For reference, typical 1-inch-thick glazing is approximately STC 30-35. Enhanced glazing with a performance of ≥STC 40 typically requires very thick double laminated or double-glazed assemblies that would add significant cost to the project. Additionally, the framing of the glass would need to be acoustically tested with the proposed enhanced glazing to ensure intended acoustical performance.

d) Interior Partitions

1. General guidance regarding interior sound-rated partitions is as follows:
   a. All partitions should extend to the structure above and be acoustically sealed to provide a good level of sound isolation.
   b. For metal studs, our preference is that all partitions be constructed with 25 gauge studs, 24” o.c. as a substantial reduction in sound transmission loss occurs when the stud gauge becomes thicker and narrower stud spacing is used.
   c. Gypsum wall and ceiling board should be type “X” with a minimum mass of 2.2 lbs/ft^2. and in lengths as long as practical to minimize the number of joints and in thicknesses indicated in the drawings.
   d. Stagger seams in configurations with multiple layers of gypsum wall board
   e. Sound attenuation batting should be minimum 2-3 lbs/ft^3
   f. Resilient channels and resilient isolation clips should be avoided, as these components are prone to installation errors degrading the sound isolation performance of a partition

g. The head and base of each partition should be sealed airtight with a non-hardening sealant.
h. Partition schedules in architectural drawings should include "Expected STC" column.
i. Penetrations due to ductwork or pipe distribution at acoustic walls should be avoided, or otherwise perimeter sealed per acoustic detailing.
j. Where cable trays occur, we strongly recommend that they do not penetrate slab to slab acoustical partitions. Cable trays should stop either side of the partition and an acoustically sealed conduit sleeve should be used at partition penetrations. Cable trays/sleeves should be located above suspended ceilings in acoustically sensitive areas.
k. Where furred or double stud partitions occur, we recommend that a 1-inch air gap be maintained between opposite rows of studs. There should be no ties between rows or between the substrate and furred wall where structurally feasible. Such notes should be clearly indicated in the drawings. Double stud partitions are also recommended between storage rooms and occupied spaces.
l. The following table provides minimum STC ratings for wood stud partitions.

STC-rated wood stud partitions					
Stud Thickness	Minimum STC 40	Minimum STC 45	Minimum STC 50	Minimum STC 55	Minimum STC 60
2x4	2+3 GYP	1+1 GYP Resilient channel	1+2 GYP Resilient channel	2+2 GYP Resilient channel	
2x6	2+3 GYP	1+1 GYP Resilient channel	1+2 GYP Resilient channel	2+2 GYP Resilient channel	
All wood stud partitions assume the following unless otherwise noted:		Isolation Clips	1+1 GYP	1+2 GYP	

<table>
<tr>
<td rowspan="2">• Full height walls extending from floor to deck
• Studs spaced 16-inches on center
• 5/8-inch Type X gypsum board
• Batt insulation with minimum 2 – 3 pcf density
Resilient channel basis of design is Clark Dietrich RC-Deluxe
Isolation clip basis of design is Pac Intl RSIC-1 and Pliteq GenieClip with 7/8” (18mil) hat channel</td>
<td>Staggered Studs</td>
<td>1+1 GYP</td>
<td>2+2 GYP</td>
<td></td>
</tr>
<tr>
<td>Double Studs</td>
<td>1+1 GYP
Minimum 1” airspace</td>
<td colspan="2">1+2 GYP
Minimum 1” airspace</td>
</tr>
</table>

e) Doors

1. Door types and door hardware details will impact the overall airborne sound isolation performance of a partition. As a reasonable balance between cost and overall acoustic performance, a reduced sound isolation performance for doors is common practice. The table below outlines typical hardware strategies for sound-reducing doors (Type SR) by space type.
2. General Door Hardware Guidance:
   a. Sound isolation door hardware options are provided below. These door treatments are intended to be a cost-effective approach to maintaining partition sound isolation performance and generally achieve the design criteria with proper equipment selection.

<table>
<tr><th colspan="4">Sound reducing door assemblies</th></tr>
<tr><th>Door Type</th><th>Acoustic Requirement</th><th>Door Hardware</th><th>Recommended Spaces</th></tr>
</table>

Sound reducing door assemblies			
SR-1	18-gauge hollow steel, fiberglass filled leaf 1-3/4in thick (min); Door lites ½in laminated glass (where applicable); Steel frame (16-gauge, grout or fiberglass filled); Door leaf must be sealed air tight	Continuous head and jamb bulb-type seals (Pemko S88 or equivalent); Astragal at double-door meeting stile (Pemko 379R or equivalent) Surface-mount, mortised, or semi-mortised automatic threshold drop seal (Pemko 411PKL, 4131SL or equivalent). Rabbeted threshold with continuous caulking/adhesive underneath (Pemko 2006 or equivalent)	Mechanical, Electrical, ITC
SR-2	Solid-core timber leaf 1-3/4in thick (min) Door lites, ½-in laminated glass; Fully caulked solid wood or steel frame (16-gauge, grout or fiberglass filled) with rebates	Continuous head and jamb bulb seals around perimeter (Pemko S88 or equivalent); Astragal at double-door meeting style (Pemko 379R or equivalent); Surface-mount, mortised, or semi-mortised automatic threshold drop seal (Pemko 411PKL, 4131SL or equivalent); Flat, smooth threshold with continuous caulking/adhesive underneath	Condominiums, Private Offices

Sound reducing door assemblies			
SR-3	Specialty manufactured acoustic door set Door, frame and latching hardware supplied together, assembled Designed to meet STC 50 Additional non-standard hardware may be required	Full perimeter seals per manufacturer's details Floor track per manufacturer's details	High noise equipment rooms near noise sensitive areas

f) Penetrations from Ductwork, Piping and Conduit
   1. Penetration sealing details are applicable even where partitions are not subject to a fire-rating requirement. In fire-rated partitions, the fire-sealing detail is comparable to acoustic sealing methods and should be reviewed to ensure proper acoustic seal is achieved.
   2. All penetrations, such as for duct, pipe and conduit, should have a 1/8 inch gap minimum between the penetrating service line and the partition and be acoustically sealed with a resilient caulking.

g) Partition-Façade Intersections
   1. At Façade/Interior wall intersections where demising partitions meet the façade, specific detailing is required to avoid sound flanking transmission and maintain the achievable sound isolation performance of the wall partition.
   2. Interior partition/façade intersection details will be coordinated as the design develops

5) Room Acoustics

a) The room acoustics is influenced by the room geometry, volume, finishes, and internal elements such as occupants, furniture, etc. The room acoustics is mostly analyzed in terms of reverberation time and/or average sound absorption coefficients. It is important to control the reverberation time for occupants to:
   1. Understand natural (unamplified) speech
   2. Understand reinforced speech through the installed sound reinforcement equipment
   3. Limit the magnification of mechanical and equipment noise due to excessive reverberation

b) The reduction of room reverberance is mostly achieved by the use of sound absorbing materials. Strategic architectural integration of acoustic finishes will be designed to provide acoustic control. Recommendations for location and performance will be further developed in the next design phase. Sound absorbing finishes should typically have a minimum NRC 0.70.

c) The following table outlines recommendations for acoustical finish coverages for walls and ceilings within various space types by room type.

Sound Absorption			
Space	Ceiling Acoustic Design Criteria (NRC)	Wall Acoustic Design Criteria (NRC)	Area of Coverage
Private Offices	≥0.75	-	≥90% ceiling
Lobbies, Reception, Commons	≥0.75	-	≥80% ceiling
Restaurants, Bars	≥0.70	-	≥65% ceiling
Fitness Rooms	≥0.70	-	≥60% ceiling
Residential Amenities	≥0.70	-	≥60% ceiling

d) Coordination Items – MEP

e) General guidance and best practice approaches to typical MEP systems are provided below for consideration by the MEP engineer. Additional recommendations will be given as the design progresses and equipment selections are made.

f) Building Envelope Sound Isolation below Rooftop MEP Equipment

1. Further review of the roof construction of the building will be evaluated to determine airborne sound isolation performance of the proposed envelope construction to spaces above and below the planned locations for rooftop and interior MEP equipment.

g) Background Sound Level Design Targets

We recommend the following HVAC noise levels for various spaces:

Background Sound Design Criteria		
Space	Background Sound Level Design Target (NC)	
Condominiums	Off	NC 25 - 30
	Low	NC 30 - 35
	Med	NC 35 - 40
	High	NC 40 - 45
Residential Amenities	NC 30 - 35	
Private Offices	NC 35 - 40	
Common Areas, Public Corridors	NC 40 - 45	

h) Duct Velocities

1. Ductwork will be reviewed as the design progresses. Preliminary ductwork and shaft sizes should be determined based on the following maximum duct velocities. Slightly higher velocities may be acceptable if smooth transitions/airflow can be achieved.
2. Please note that the values provided in the table below are guidelines only and assume good airflow conditions. Presence of elbows, fittings, or abrupt duct transitions may require air to run at lower velocities. Also, please note that multiple guidelines may apply to a given section of ductwork (e.g., main branch ductwork within 10-feet of a supply terminal). In such cases, we recommend that ductwork is sized according to the most stringent applicable guideline. In spaces with "-" we recommend that the indicated ductwork condition does not occur in spaces of the indicated acoustical sensitivity.
3. We recommend the following to reduce airflow-induced noise:

d) Provide smooth duct turns, transition fittings, and simple duct layouts

e) Wherever possible, avoid installing elbows, tees, take-offs, and other fittings adjacent to each other.

f) Provide a minimum of 3-duct diameters of straight duct between fittings.

g) For flexible ductwork, provide at final connections to diffusers with smooth connection and no kinks.

h) Select the system to perform at the lowest applicable fan speed with dampers open.

i) In general, we recommend that duct dimension ratios should not exceed 3:1.

Maximum allowable duct velocities						
Noise Criterion		NC 45	NC 40	NC 35	NC 30	NC 25
Main branch above suspended ceiling		2000	1800	1500	1300	-
Main branch in exposed configuration		1600	1500	1200	1100	-
Duct within 10 to 20 feet of supply diffuser/return grille	Supply	900	850	800	700	550
	Return	1000	950	900	800	650
Duct within 0 to 10 feet of supply diffuser/return grille	Supply	700	650	600	500	450
	Return	800	750	700	600	500
Supply diffuser – 'free velocity'		550	500	450	400	300
Return grille – 'free velocity'		650	600	550	500	400

Open return duct above ceiling	850	750	650	-	-
Open return duct exposed	800	700	600	-	-

i) Terminal Units
1. Locate volume control boxes away from noise-sensitive rooms (e.g., more stringent than NC 35) to prevent radiated noise breakout to the room. As the BOD manufacturer is finalized, volume control box selections are being actively coordinated internally to ensure design target NC levels are not exceeded.
2. Where continuous finished ceilings occur, we recommend that volume control boxes are selected with radiated and discharge noise ratings of at least 10 NC points below the background noise criterion of the space above which the units are located and the space served. Where exposed ceilings or non-continuous ceiling finishes occur, radiated noise ratings should be at least 15 NC points below the background noise criterion of the space above which the units are located.

j) Diffusers & Grilles
1. Diffusers should be selected such that they are 10-NC points less than the space serviced (e.g., Private offices should incorporate diffusers selected for a maximum of NC-25).

k) Internal Duct Lining
1. Acoustical duct liner should be composed of fiberglass or cotton fiber of 3-5 pcf density and a minimum rating of NRC 0.80.
2. Supply, return and exhaust air ductwork from air handler units will be lined with 1-inch-thick internal duct liner for the first 25 feet from the units.
3. Incorporate a minimum of 10 feet of duct liner after any discharge side of VAV boxes. Additional duct liner may be required for spaces rated ≤NC 30.
4. The feasibility of incorporating internal duct lining and/or sound attenuators shall be reviewed with the owner and mechanical engineer. If fiberglass lining is not allowed, custom units which achieve lower supply, return, and/or casing radiated noise levels should be considered for costing purposes.

l) Balancing Dampers
1. Noise is generated as airflow passes through balancing dampers, typically perceived as a whistle or hiss.
2. To mitigate this noise, we recommend that a minimum of 5 feet of acoustical flex duct or internally lined sheet metal duct with a minimum lining thickness of 1 inch is included between diffusers and balancing dampers. It is very important that

dampers are not incorporated into grilles. The use of diffuser blades for "balancing" the air system should be avoided.

m) Acoustical Flexible Ductwork

1. Acoustical flexible ductwork should be installed per manufacturer specifications. If flex duct is kinked in any way, which results in turbulent airflow, it should be replaced or reconfigured. The length should not exceed 5 feet to limit sagging and damage. We recommend that an acoustical flex duct similar to Flexmaster 8M be used.

n) Return Air Transfers

1. In locations where full height partitions are scheduled and returns are not fully ducted, transfers for air will be required to meet return air requirements set by the mechanical engineer. In general, transfers should occur through 1-inch internally lined U or Z-shaped transfer ducts sized for a maximum 500 fpm. As previously mentioned, the mechanical engineer shall confirm whether internal duct lining is allowable. If not, alternate sound attenuation measures at return air transfers shall be coordinated.
2. The height dimension of the duct after the 90-degree bends should be equivalent to minimum 2 width dimensions of the duct.
3. The return transfer should occur at the vertical face above the entry door and not in demising walls of enclosed rooms. As the mechanical design is developed, it should be kept in mind that there should be no transfers that occur between noise sensitive rooms

o) Duct Layout and Crosstalk

1. Duct runs in areas with slab-to-slab partitions should be configured to maximize separation between takeoffs to individual rooms in order to reduce duct borne room-to-room crosstalk. Duct takeoffs to terminal grilles should be staggered to maximize the available acoustical separation provided.
2. Ductwork should be routed to avoid clashing with slab-to-slab partitions. Where this condition cannot be avoided, partitions should be jogged around the ductwork such that the ductwork remains on the less acoustically sensitive side of the partition.

p) Electrical Junction Boxes

1. Outlets should not be installed back-to-back, but instead in separate stud cavities with a minimum separation of 24 inches. For devices recessed into demising walls, we also recommend that the junction boxes be wrapped in putty pads similar to Hilti Type CP-617.

q) Vibration Isolation – Mechanical Equipment

1. All major mechanical equipment will require vibration isolation per the guidelines set out in ASHRAE 2019 HVAC Handbook: Chapter 49 "Sound and Vibration Control". In some cases, isolation methods may be employed which exceed ASHRAE guidance in order to achieve lower noise or vibration levels. Special attention should be given to rooftop equipment in order to attenuate vibration transmission. Further recommendations will be issued in later design stages after the building structural and mechanical strategies have been confirmed.
2. For costing and design purposes, all operating mechanical and electrical equipment and attached pipe shall be vibration isolated in accordance with the guidelines in the 2019 ASHRAE Handbook – HVAC Applications, Chapter 49 - Noise & Vibration Control, Table 47.
3. In some cases, isolation methods may be employed which exceed ASHRAE guidelines in order to achieve noise or vibration levels that meet the design targets.
4. All large equipment should be located on stiff and massive slabs or rooftop dunnage steel (if any). No equipment should be mounted to lightweight structure.
5. Pipework should be supported on isolation hangers throughout the equipment rooms, and for at least 50 feet from the connected equipment. In some cases, additional isolation may be warranted.
6. All electrical, pipe and duct connections to equipment should be via flexible connections.
7. Large diameter chilled water pipes should not be supported from structure common to the acoustically critical areas. These large, noisy pipes should run only in spaces that are inherently noisy (major equipment rooms, loading dock, etc.). Vibration isolators alone do not provide adequate protection against the high vibration energy from such pipes.

r) Plumbing Pipe Installations

1. Noise and vibration control measures should be implemented into the design, including resilient support for plumbing lines, flexible pipe work at connections to vibration isolated equipment, selection of moderate plumbing fluid velocities and vibration isolation for the pumps.
2. Plumbing pipes located in double stud partitions should be mounted on the service side studs only and should not have any rigid connection to the opposite row of studs.

s) Electrical Systems

1. The electrical system contains various components that create noise. Unlike noise from the HVAC system, which is typically broadband in nature, electrical components tend to generate highly tonal noise. Such noise can be annoying or distracting even at moderate noise levels. As such, noise and vibration control measures should be incorporated into the design of the electrical system. Typically, these would include the use of proprietary vibration isolators for sub-stations, transformers and some distribution panels. Detailed recommendations and specifications will be developed as the project progresses. Further review of noise impact to the environment and to the interior of the building from the proposed emergency generator will be assessed as the project progresses.

6. Energy Compliance Requirements

a) Introduction

1. This project is governed by the 2021 International Energy Conservation Code with Montana Amendments. Montana allows a project to either comply with 2021 IECC or completely switch to demonstrate compliance via ASHRAE 90.1 2019, which is the chosen path for this project.
2. Pending approval from the City, the project will demonstrate energy code compliance using either (a) a single whole-site energy model, or (b) a whole-site prescriptive compliance approach. The primary basis for this request is the shared parking garage/podium (containing some fully conditioned spaces), which creates practical difficulties in demonstrating energy code compliance as a stand-alone permit.
3. While this portion of the narrative identifies key energy code compliance items, the specific design elements are still included within the respective trade narrative sections. For example, if an onsite photovoltaic system is required as part of the energy code compliance package, that requirement will be indicated in this section, but the system capacity, components, and design elements will be described in detail in the Electrical Design Requirements.

b) General Project Energy Code Compliance Path

1. This project will utilize the Energy Cost Budget Method to demonstrate energy compliance, as described in ASHRAE 90.1

2019. ASHRAE 90.1 Sections X.4 list all of the mandatory compliance measures that CANNOT be traded off in the energy model.

2. The ECB analysis and documentation are by IMEG.
3. At the time of building permit complete, stamped, architectural, mechanical, electrical, and plumbing sets need to be submitted for review by the AHJ, unless deferred submittals have been specifically coordinated

c) Specific Energy Code Compliance Notes

1. This section does not list every required item in the energy code, but instead provides details on sections of the energy code that may require significant explanation or coordination. The design specifics for complying with these items are included in the respective Architecture, HVAC, Plumbing, and Electrical drawings and narratives.

d) ASHRAE 90.1 2019 Section 5 Building Envelope

1. Because this project is utilizing the Energy Cost Budget compliance path, there is no requirement for the envelope to perform better than the prescriptive values, however it is recommended to minimize heat loss throughout the building to reduce energy usage. Assembly performance targets are listed in Section 11 below.
2. Thermal envelope, moisture barrier, and air barrier design are by architect and/or building envelope and air barrier consultant.
3. IMEG will maintain the envelope calculation and coordinate thermal envelope assembly performance requirements with design team.

e) ASHRAE 90.1 2019 Sections 6-9

1. The Mechanical/Plumbing/Electrical/Lighting Contractor(s) shall be responsible for general compliance with the mandatory requirements of Sections 6 – 9
2. ASHRAE 90.1 2019 Section 11 Energy Cost Budget Introduction
3. This project will follow the energy code compliance path defined in ASHRAE 90.1 2019 Section 11 Energy Cost Budget, which is a whole-building energy model and allows for trade-offs between envelope, mechanical system, electrical, and service hot water system efficiencies. The Energy Conservation Measures (ECMs)

presented in the subsection below work together to meet the code required building performance and carbon emissions metrics.

f) Section 11 ECB Energy Conservation Measures Descriptions

1. Glazing:
    a. 40% maximum glazing percentage (ratio of “punched window opening area” to “gross vertical above-grade walls bordering conditioned space”)
    b. Calculation includes walls at conditioned spaces in garage
    c. Portions of the project can exceed 40% glazing percentage however, and it is anticipated that the hotel portion may have the highest glazing ratio.
2. High-performance dual pane vinyl glazing
    a. Approx. U-0.26 target U-Factor for fixed glazing
    b. Approx. U-0.28 target U-Factor for operable glazing
3. High-performance window wall glazing
    a. Approx. U-0.33 target U-Factor for fixed glazing
    b. Approx. U-0.37 target U-Factor for operable glazing

4. Opaque Envelope:
    a. Minimize uninsulated assemblies in thermal envelope
    b. Steel frame walls cavity insulation plus 2” Continuous insulation with thermally broken attachment system
    c. Slab on grade will be 2’ of vertically installed insulation
    d. Garage Walls for heated or cooled spaces – Maximize wall insulation, limited opportunity for under-insulated walls, no opportunity for uninsulated walls in thermal envelope (water entry, fire pump, etc.)
    e. Roof U < 0.03 (min R-30 c.i.)
    f. Floor U < 0.03 (min R-30 pinned insulation)

g) Section 5.4.3 Air Leakage
1. Must pass whole building air leakage test at 0.40 cfm/sf or less
2. Dwelling unit Heating and Cooling: Single zone split systems
3. Alternate: Multi zone split system
4. Amenity HVAC: AS VRF with Dedicated Outdoor Air System with ERV
5. Corridor Ventilation: Air source heat pump w/ ducted return and economizer capabilities

6. Dwelling unit Ventilation:
7. Unit by Unit ERV system; > 60% Overall ERV Effectiveness
8. Domestic hot water provided by natural gas condensing boilers w >95% thermal efficiency

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Appendix A
Terms, Definitions, Glossary

A. General

B. Energy Modeling

C. Structural

D. Civil

E. Mechanical

F. Plumbing

G. Electrical

H. Architectural Lighting

<u>0–10 V</u>

1. DC voltage control signal varies between zero and ten volts.
2. <u>Color Rendering Index (CRI)</u>
   A metric used to describe how faithfully a light source can render the true colors of objects and spaces.
3. <u>Correlated Color Temperature (CCT)</u>
   a. The appearance of light in terms of warmth or coolness
   b. The Kelvin range is typically broken down as follows: Warm – 0 – 2900K; Bright White – 3000K – 4900K; Cool Daylight – 5000K – and above.

4. <u>Lumen</u>
   Measurement unit for the lighting output of lamps or fixtures. The total lumens emitted and their spatial distribution are of paramount importance when creating appealing and luxurious indoor spaces. In lighting, lumens can be compared to miles traveled and watts can be compared to fuel consumption.
5. <u>Foot-candle (fc)</u>
   Measurement unit for illuminance, or lumens per unit of area. One foot-candle is equivalent to one lumen per square foot (refer to Illuminance).
6. <u>Glare</u>
   Visual impairment caused by a bright source of light, directly visible or reflected by a surface. There are two types of glare:
   a. Discomfort glare causes an instinctive reaction to close the eyes and look away.
   b. Disability glare impairs vision but does not cause the same reaction as discomfort glare.
7. <u>Recessed Down Lighting</u>

Direct luminaires that are installed into the ceiling plane, the housing sits above the ceiling and only the trim and light aperture remain visible from below, to provide ambient, accent, or task illumination. They may incorporate fixed, adjustable, or wall-wash optics, and are available in a wide range of beam spreads, color temperatures, and optical control options.

8. <u>Surface Mounted Lighting</u>

Direct luminaires that are installed directly onto the ceiling or wall. The luminaire's housing remains fully visible and sits flush against the mounting surface.

9. <u>Linear Lighting</u>
   Multiple LEDs (light-emitting diodes) aligned in a single strip and used for creating uninterrupted lines of directional lighting.
10. <u>Wall Grazing</u>
    Direct illumination positioned near surfaces at a shallow angle which emphasizes surface texture, relief, and material variation by creating pronounced highlights and shadows.

11. <u>Cove Lighting</u>
    A type of lighting that generally directs its output towards the ceiling, and where individual fixtures are hidden in ledges. Cove lighting is often used for decorative purposes because it can emphasize the borders of walls, as well as ceiling features.

APPENDIX B

(Attach LEED Checklist if applicable)

APPENDIX C

A. Space Design Requirements [This can be quite time consuming and can be omitted for most projects. For buildings where we want to document requirements for different space types, you can do that within the basis of design document. This may be helpful to document the Owner's requirements or code requirements for different spaces. Generally used in large hospitals or lab spaces with a lot of varying space requirements and when the Architect is not documenting this information in a Room Data Sheet.]

1. Space Type A

a. HVAC Requirements [Temperature, ventilation, air changes, humidity or any special requests]

b. Plumbing [Fixtures and connections required]

c. Medical Gas [Number and type]

d. Lighting Requirements [Luminaire types or lighting levels]

e. Normal Power Requirements [Number and location of outlets, including any requirements for mobile equipment]

f. Emergency Power Requirements [Number and location of outlets, including which branch of power is required]

g. Fire Alarm Requirements

h. Data/Technology Requirements [Number and location of data outlets or other low voltage system requirements]

i. Number of People [Documentation for heat gain and ventilation]

j. Equipment in Space [Documentation for power, heat gain, data, structural, or plumbing requirements]

k. Floor or Roof Loading Requirements

l. Acoustical Considerations

2. Space Type B

   a. HVAC Requirements [Temperature, ventilation, air changes, humidity or any special requests]

   b. Plumbing [Fixtures and connections required]

   c. Medical Gas [Number and type]

   d. Lighting Requirements [Luminaire types or lighting levels]

   e. Normal Power Requirements [Number and location of outlets, including any requirements for mobile equipment]

   f. Emergency Power Requirements [Number and location of outlets, including which branch of power is required]

   g. Fire Alarm Requirements

   h. Data/Technology Requirements [Number and location of data outlets or other low voltage system requirements]

   i. Number of People [Documentation for heat gain and ventilation]

   j. Equipment in Space [Documentation for power, heat gain, data, structural, or plumbing requirements]

   k. Floor or Roof Loading Requirements

APPENDIX D
STRUCTURAL DESIGN CRITERIA

A. Seismic Load Criteria
   1. Site Soil Classification
      a. Risk Category
      b. Seismic Design Category
      c. Seismic Force Resisting System

B. Foundation Criteria
      a. Allowable Net Soil Bearing Pressure

C. Structural Steel Criteria
      a. Material Properties

Wide flange beams	ASTM A992 (Gr. 50)
Wide flange columns	ASTM A992 (Gr. 50)
Channel sections	ASTM A992 (Gr.50)
Angle sections	ASTM A572 (Gr. 50)
Hollow structural sections (HSS)	ASTM A500, Gr. C
Pipe steel	ASTM A53, Gr. B
Plates	ASTM A572 (Gr. 50)
Anchor rods	ASTM F1554, Gr 36 55

D. Concrete Criteria
      a. Material Properties

Reinforcing steel	ASTM A615, Grade 60
Welded wire reinforcement	ASTM A1064
Fiber reinforcement	ASTM C1116

B. Strength Properties

Drilled piers & footings	3000 psi
Slab on deck	3000 3500 psi
Slab on Grade	3000 psi
Columns	5000 psi
Structural slabs/beams	5000 psi
All other concrete	4000 psi

E. Masonry Criteria
      a. Material properties:

Concrete masonry units	ASTM C90
Mortar materials	ASTM C270, Type S N
Grout	ASTM C476
Joint reinforcing	ASTM A82

d. Strength properties:

Design assembly strength, f'm	2000 2500 3000 psi
Individual concrete masonry units	2000 3250 4500 psi
Grout	2000 2500 3000 psi

F. Cold-Formed Criteria

e. Material properties:

18 gauge and thinner	ASTM A653, Gr 33
16 gauge and thicker	ASTM A653, Gr 50
Connection material (> 3/16")	ASTM A36
Electrodes for arc welding	AWS 5.1, E60XX

f. Coatings:

Hot-dip	ASTM A924, G60
Electro-plate	ASTM A591
Aluminum-zinc	ASTM A792, Gr 40